DFA:

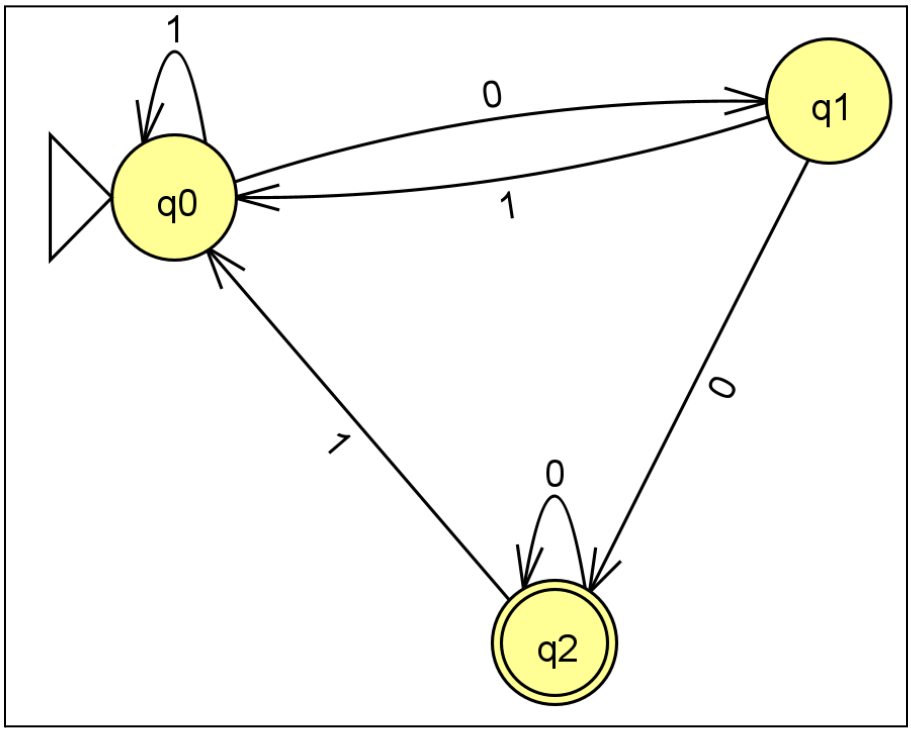
Question 1: Design a DFA for the language accepting strings ending with '00' over input alphabets {0, 1}

Solution:

Language:

L = {00, 100, 1100, 111100, …}

State Transition Diagram:



Testing various Inputs:

Input	Result
00	Accept
11	Reject
111100	Accept
000010	Reject
11001	Reject
101010	Reject
	Reject

State Transition Function:

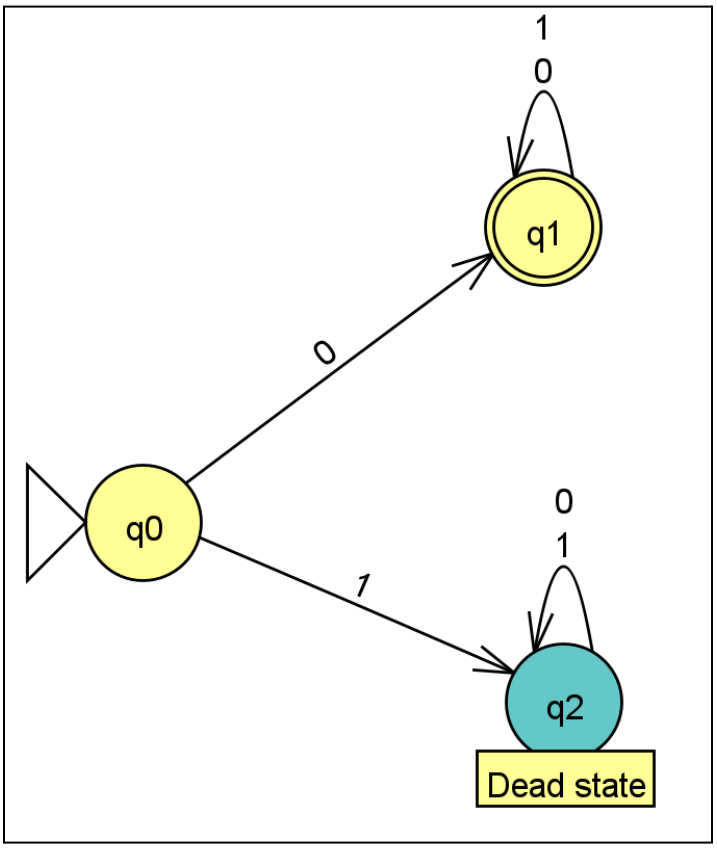
State vs Input	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_0

Question 2: Design a DFA that accepts all strings beginning with '0' over Σ= {0,1}

Solution:

Language = 0, 01, 0001, 010, 01111,…}

State Transition Diagram:



Testing various Inputs:

Input	Result
0	Accept
1	Reject
01	Accept
100	Reject
10000	Reject
000	Accept
001	Accept
00002	Reject

State Transition Function:

State vs Input	0	1
q_0	q_1	q_2
q_1	q_1	q_1
q_2	q_2	q_2

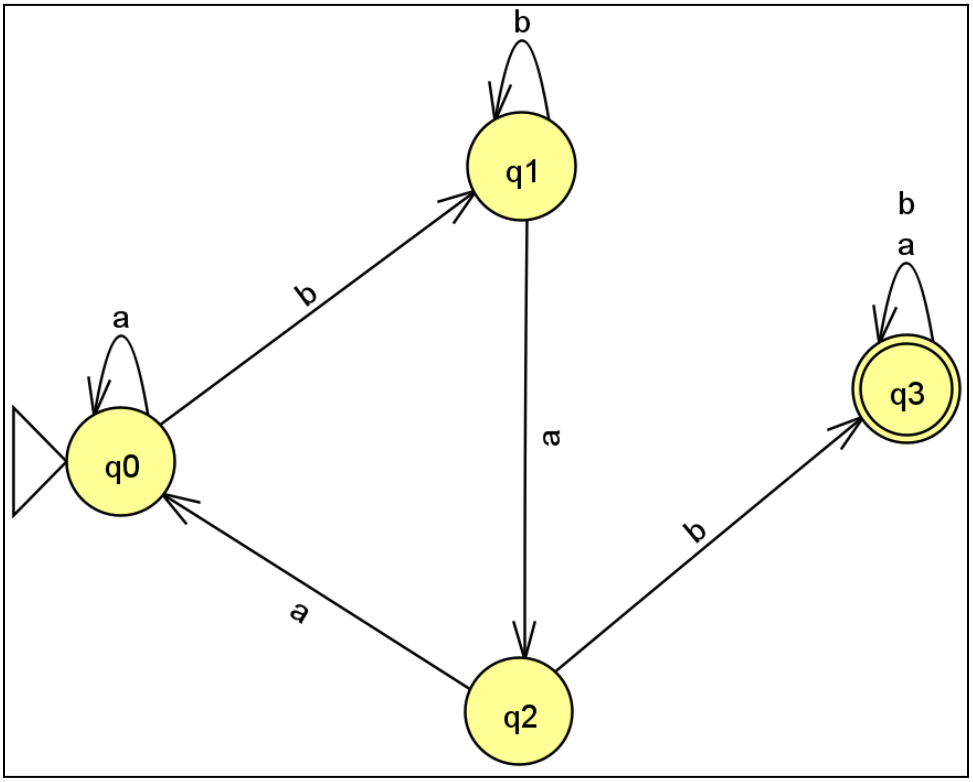
Question 3: Design a DFA that accepts all strings having substrings 'bab' over ∑ = {a,b}

Solution:

Language:

L = {bab, ababb, bbab, abab, baba, babba}

State Transition Diagram:



Testing various Inputs:

Input	Result
bab	Accept
aba	Reject
baba	Accept
bbbb	Reject
aaaa	Reject
23	Reject

State Transition Function:

State vs Input	a	b
q_0	q_0	q_1
q_1	q_2	q_1
q_2	q_0	q_3
q_3	q_3	q_3

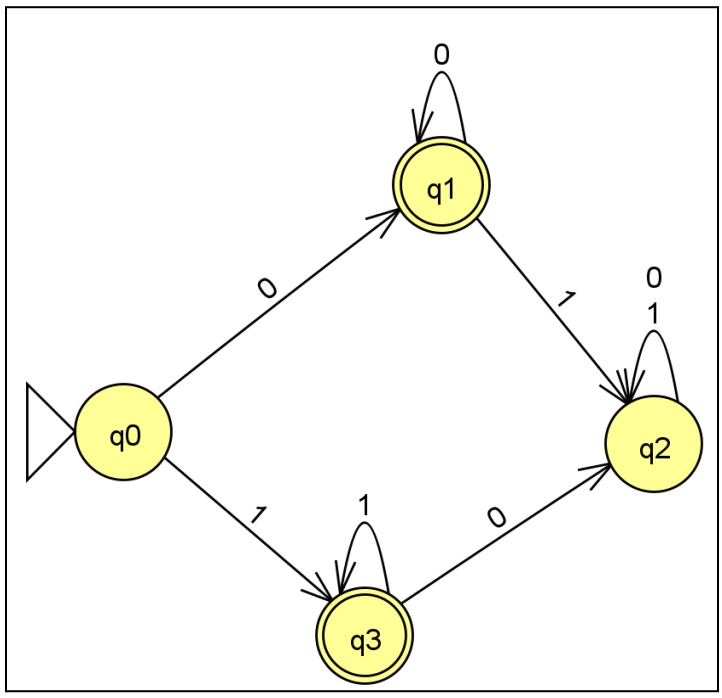
Question 4: Design a DFA for the language accepting strings with '0' and '1' only over input alphabets {0, 1}

Solution:

Language:

L = {0, 1, 00, 11, 000, 1111, …}

State Transition Diagram:



Testing various Inputs:

Input	Result
0	Accept
1	Accept
00	Accept
11	Accept
0011	Reject
01	Reject
10	Reject
aba	Reject

State Transition Function:

State vs Input	0	1
q_0	q_1	q_3
q_1	q_1	q_2
q_2	q_2	q_2
q_3	q_2	q_3

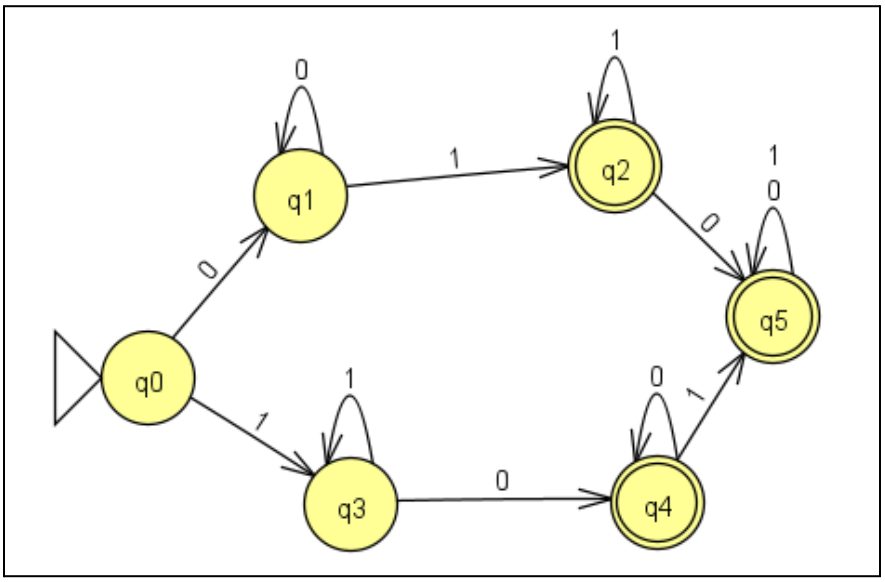
Question 5: Design a DFA for the language accepting strings containing '01' or '10' as a substring over input alphabets Σ= {0, 1}

Solution:

Language:

L = {10, 01, 100, 010, …}

State Transition Diagram:



Testing various Inputs:

Input	Result
01	Accept
10	Accept
000	Reject
111	Reject
1000	Accept
0010	Accept
11101	Accept

State Transition Function:

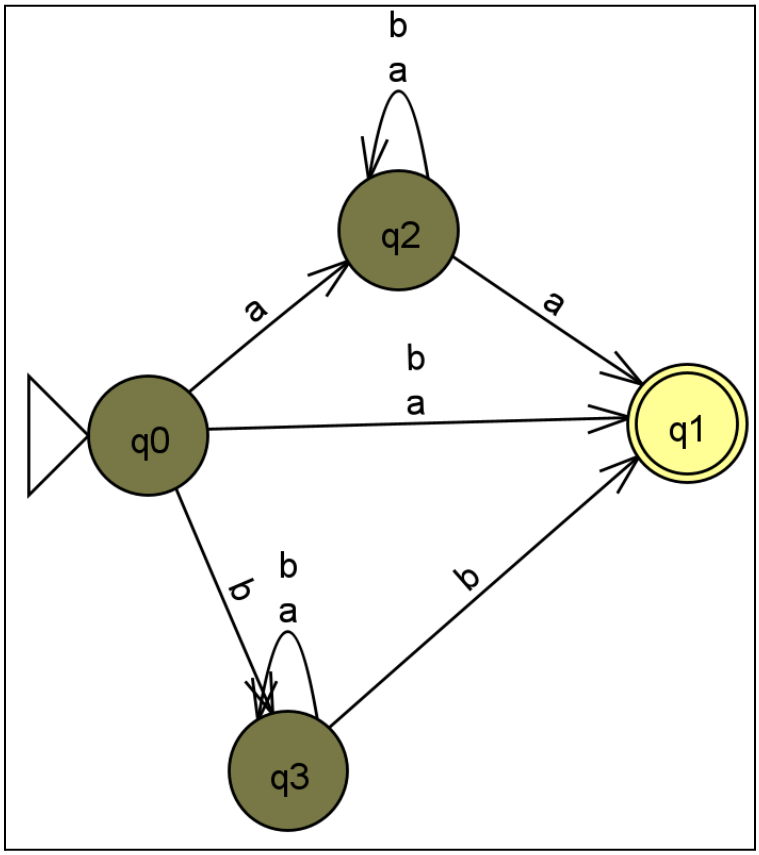
State vs Input	0	1
q_0	q_1	q_3
q_1	q_1	q_2
q_2	q_5	q_2
q_3	q_4	q_3
q_4	q_4	q_5
q_5	q_5	q_5

NFA:

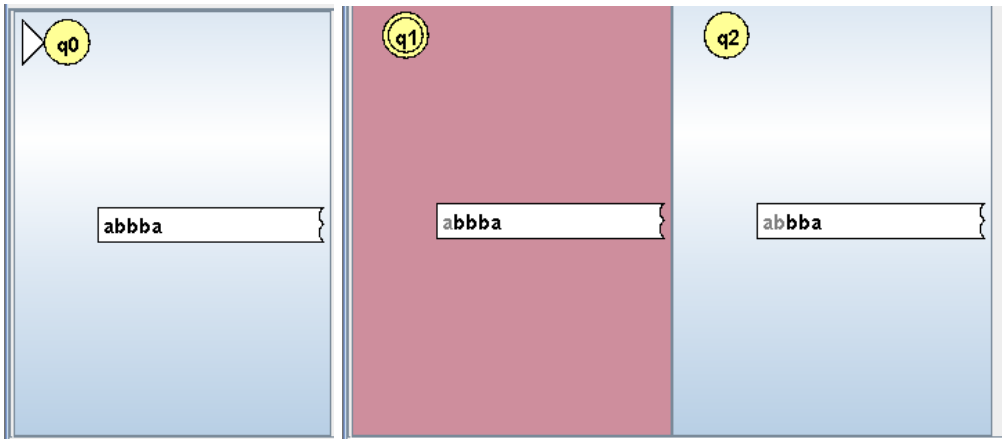
Question 6: Design a NFA that accepts all strings over alphabet Σ= {a, b} such that every accepted string start and end with same symbol

Solution:

State Transition Diagram:



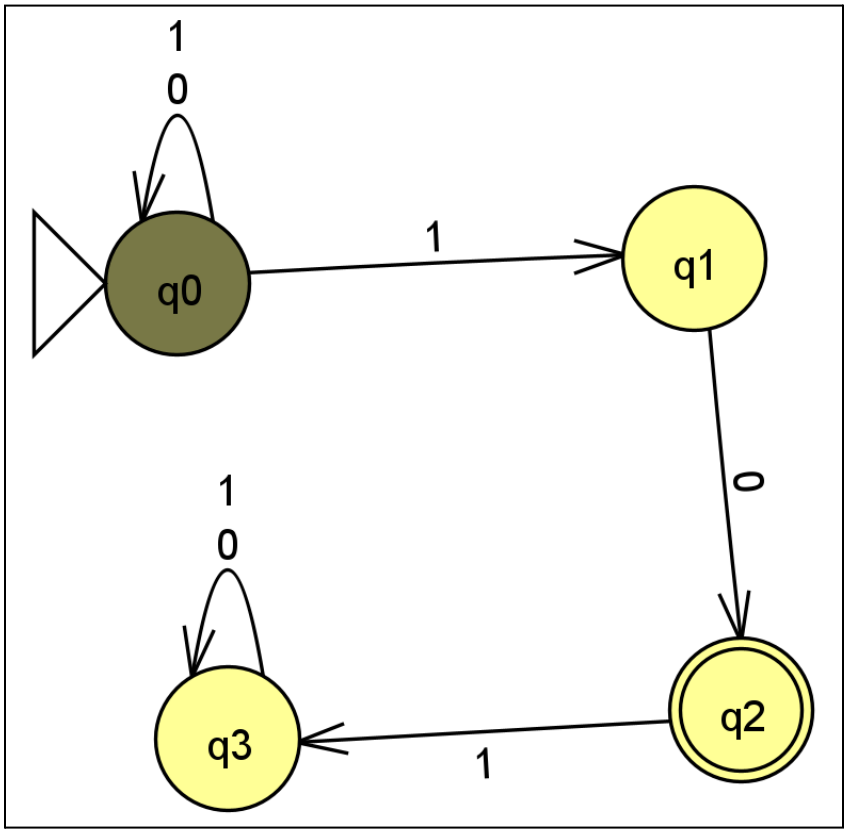
Testing Input:



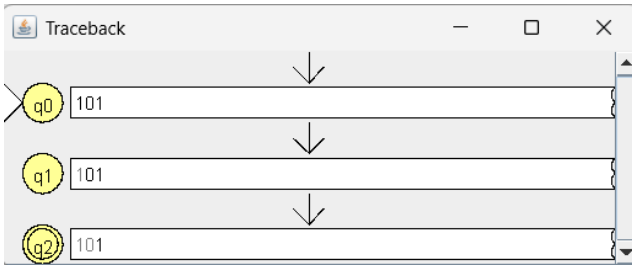
Question 7: Design an NFA to recognize the language consisting of all strings containing one or more instances of '101' over alphabet Σ= {0, 1}

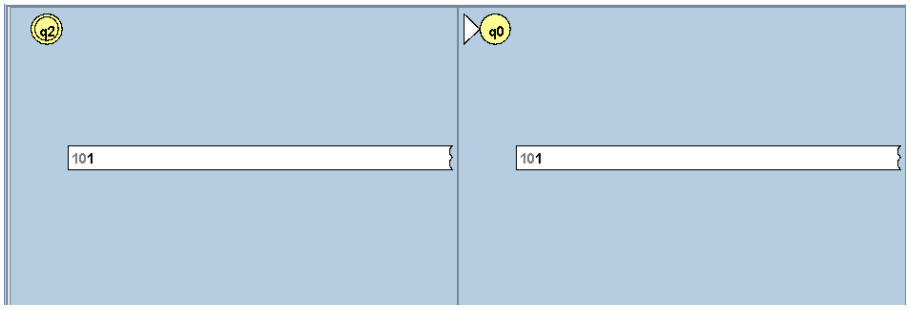
Solution:

State Transition Diagram:

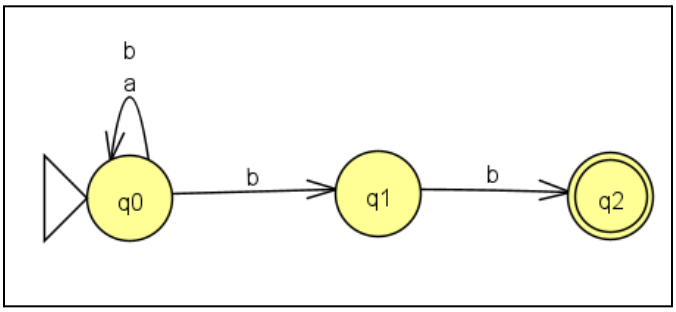


Testing Input:





Question 8: Convert given NFA to DFA

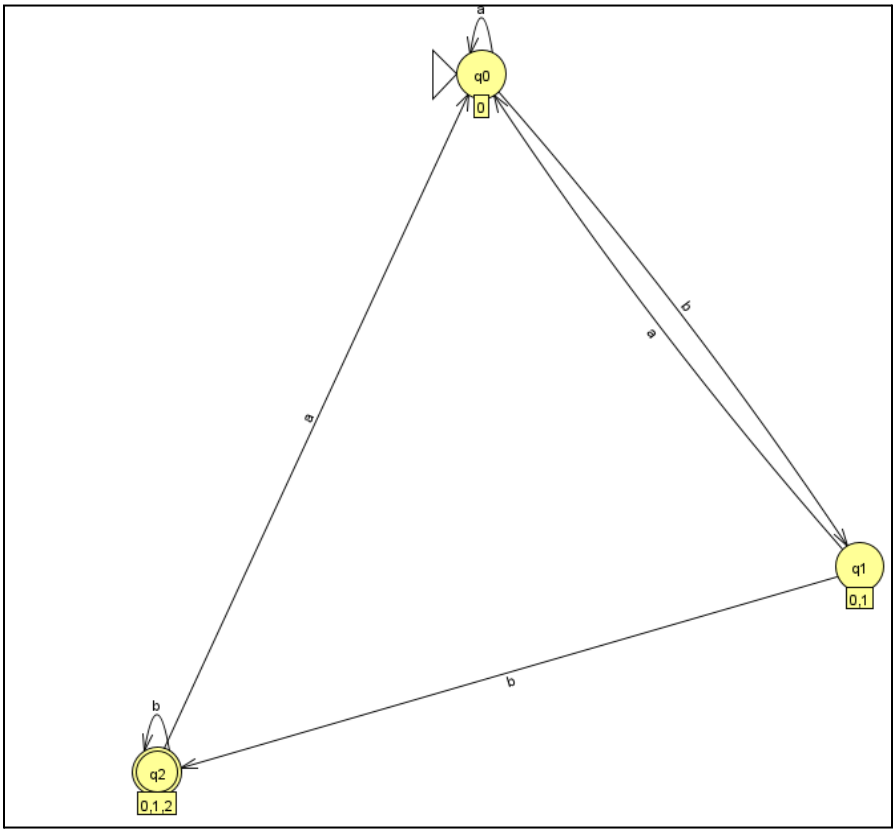


Solution:

Transition Table for NFA:

State vs Input	a	b
q_0	q_0	q_0 , q_1
q_1	-	q_2
q_2	-	-

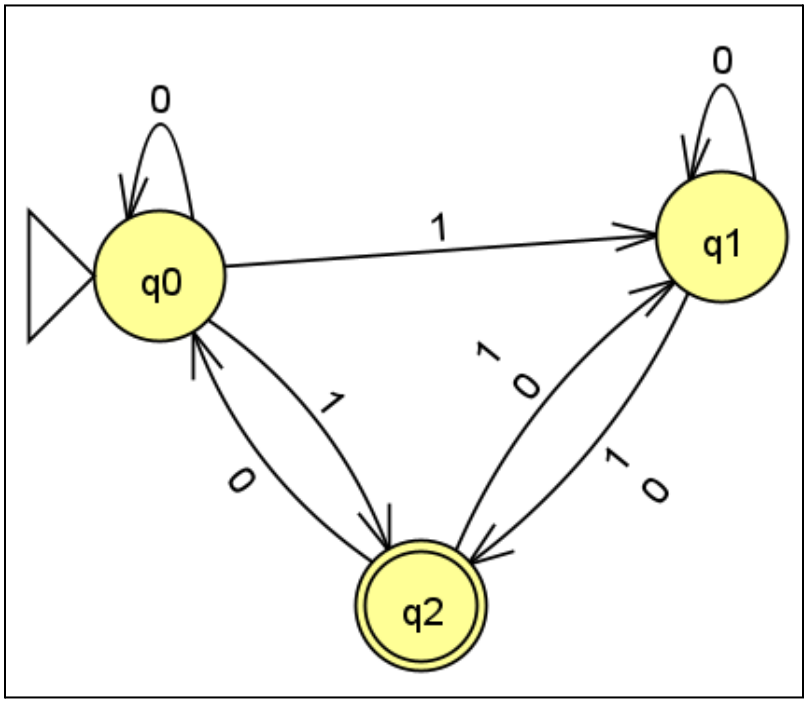
State Transition Diagram DFA:



Transition Table for DFA:

State vs Input	a	b
→ q_0	q_0	$\{q_0 , q_1\}$
$\{q_0 , q_1\}$	q_0	$*\{q_0 , q_1 , q_2\}$
$\{q_0 , q_1 , q_2\}$	q_0	$*\{q_0 , q_1 , q_2\}$

Question 9: Convert given NFA to DFA



Solution:

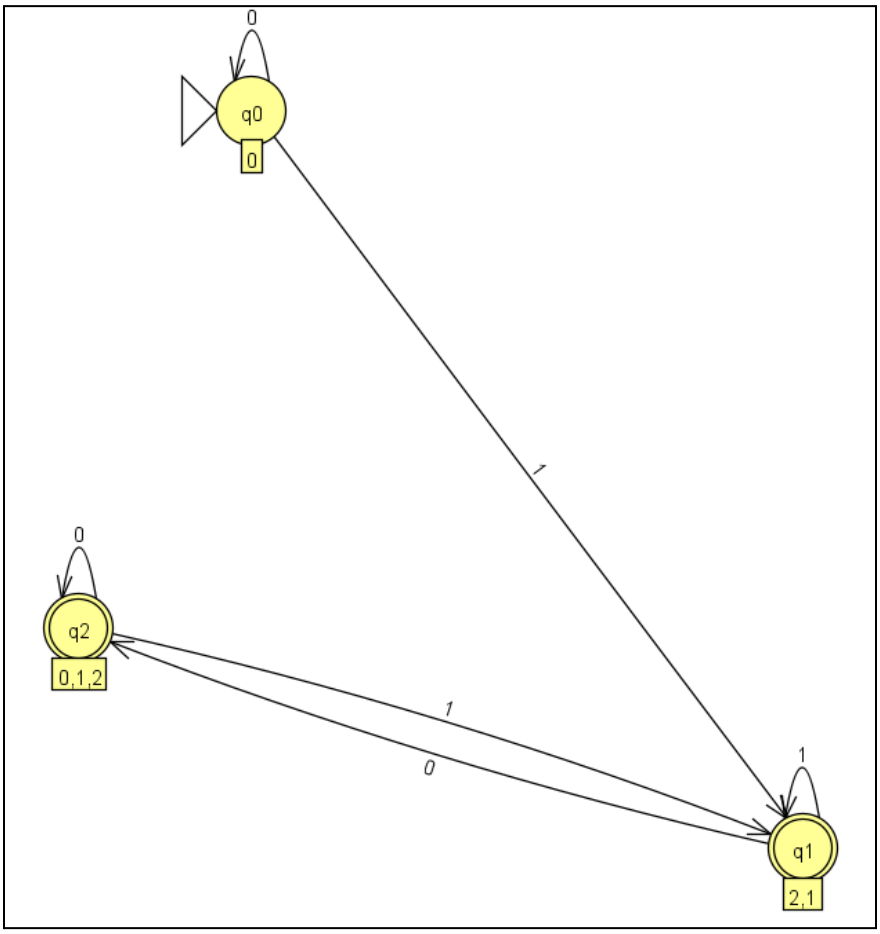
Transition Table for NFA:

State vs Input	0	1
q_0	q_0	$\{q_1 , q_2\}$
q_1	$\{q_1 , q_2\}$	$*q_2$
q_2	$\{q_0 , q_1\}$	q_1

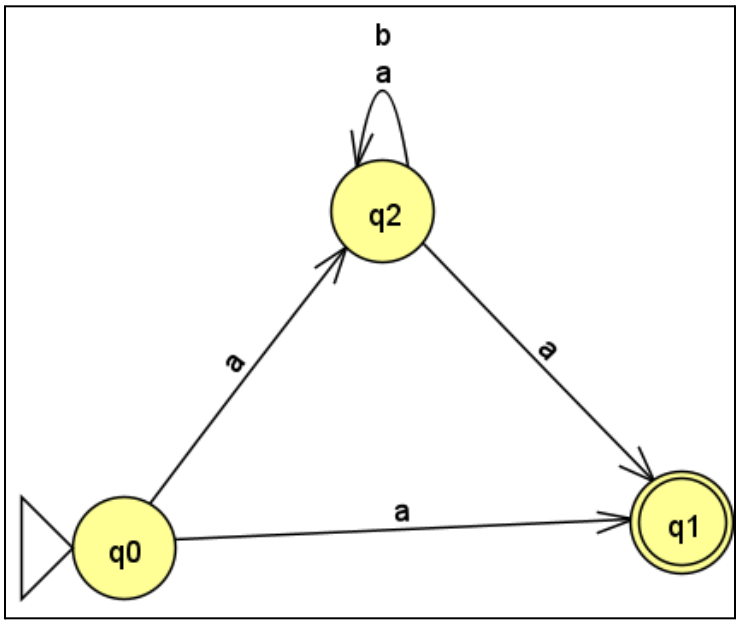
Transition Table for DFA:

State vs Input	a	b
$\rightarrow q_0$	q_0	$\{q_1 , q_2\}$
$\{q_1 , q_2\}$	$*\{q_0 , q_1 , q_2\}$	$\{q_1 , q_2\}$
$*\{q_0 , q_1 , q_2\}$	$*\{q_0 , q_1 , q_2\}$	$\{q_1 , q_2\}$

State Transition Diagram DFA:



Question 10: Convert given NFA to DFA



Solution:

Transition Table for NFA:

State vs Input	a	b
q_0	$\{q_1, q_2\}$	
q_1		
q_2	$\{q_1, q_2\}$	q_2

Transition Table for DFA: (ϕ: Dead State)

State vs Input	a	b
$\rightarrow q_0$	$*\{q_1, q_2\}$	ϕ
$*\{q_1, q_2\}$	$*\{q_1, q_2\}$	$\{q_2\}$
q_2	$*\{q_1, q_2\}$	$\{q_2\}$
ϕ	ϕ	ϕ

State Transition Diagram DFA:

